Software Requirements

Assignment 1

Team: Dev Core
UBB-SE-202

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Administrator

1. Manage shift and guard schedules - as an administrator, I need to handle the scheduling of shifts, including ER guards and pharmacy rotations. I can do this by navigating to the Shift Scheduling dashboard. I need to be able to:
    - Create new shifts by pressing a "Create Shift" button, which allows me to assign a specific doctor to an ER guard shift or schedule a pharmacist for work. The system must filter the available staff based on the specialization (for Doctors) or certification (for Pharmacists) required for the shift's location.
    - Generate and display daily or weekly shift rosters that visually show who is on duty in specific departments.
    - Modify shift statuses (e.g., changing a shift from "Scheduled" to "Active") or reassign a shift to a different staff member by selecting an existing shift block. When a shift is set to "Active," the system must automatically set the staff member's is_available status to True.
    - Cancel a scheduled shift by selecting it and pressing a "Cancel Shift" button. Canceling an "Active" shift must immediately update the staff member's is_available status to False and their DoctorStatus to OFF_DUTY.

User Action	System Flag or Database Update	Important Info and Constraints
Create New Shift	Shift.staff_id (New Record)	Specialization Check: System filters staff based on required specialization (Doctors) or certification (Pharmacists) for the shift's location.
Set Shift to Active	Shift.status -> ACTIVE	Automatically sets Staff.is_available to True.
Reassign Shift	Shift.staff_id (Update)	Integrity Check: System verifies the new staff member is not already assigned to an overlapping shift.
Cancel Active Shift	Staff.is_available -> False	Sets DoctorStatus to OFF_DUTY

2. Manage patient appointments - as an administrator, I need to manage the scheduling of standard medical appointments outside of the emergency room workflow. I can do this by navigating to the Appointments module. I need to be able to:
    - View a calendar component displaying all upcoming appointments for any selected doctor.
    - Book a new appointment by pressing "New Appointment", where I can assign a specific doctor to a patient by utilizing the patient identifier provided by the Patient Management team.
    - Update an appointment's status (e.g., from "Scheduled" to "Finished") by opening the appointment details. Updating an appointment to "Finished" must automatically trigger a check: if the doctor has no other concurrent appointments, their doctorStatus must revert from IN_EXAMINATION to AVAILABLE.
    - Cancel an existing appointment by selecting the event and pressing "Cancel Appointment". The system must validate that the status of the appointment is not already "Finished" before allowing cancellation.

User Action	System Flag or Database Update	Important Info and Constraints
Book Appointment	Appointments.patient_id	Data Source: Utilizes the patient identifier provided by the Patient Management team (external_ref_id).
Finish Appointment	Appointments.status -> Finished	Doctor State Reset: If no concurrent tasks exist, DoctorStatus reverts from IN_EXAMINATION to AVAILABLE.
Cancel Appointment	Appointments.status -> Canceled	State Validation: System blocks cancellation if the current status is already marked as "Finished".

3. Automated Fatigue Management & Shift Balancing

- As an administrator, I can run an "Auto-Audit" on the weekly shift rosters before publishing them to the staff.
- The system must validate every scheduled shift against labor laws: checking that no staff member exceeds 60 hours per week and ensuring there is a mandatory minimum 12-hour rest period between the end of one shift and the start of another. This rest period is calculated by comparing the end_time of the previous shift with the start_time of the next shift for the same staff_id.
- If a violation is detected, the system blocks the roster from being published and highlights the conflicting shifts in red.
- To resolve the conflict, the system provides an "Auto-Suggest" feature that recommends an alternative staff member with the exact same role (e.g., ER Doctor) who currently has the lowest accumulated hours for that month and is not already scheduled. The "Auto-Suggest" logic must also verify that the alternative staff member's is_available status is False at the time of the proposed shift to prevent double-booking.

System Trigger	Validation Rule	Automation / Resolution
Run Auto-Audit	Shift.start and Shift.end	Sums hours to ensure no staff_id exceeds 60 hrs/week and validates a 12-hour gap between shifts.
Conflict Detected	UI Alert (Red Highlight)	System Block: Prevents the roster from being published to the staff until the violation is resolved.
Auto-Suggest	Staff.is_available and Shifts Table	Recommends staff with the same role/specialty who are currently is_available = False and have the lowest monthly hours.

4. Automated ER Dispatch & Doctor Status Management

- Requirement: As an administrator managing the Emergency Room workflow, I need the system to automate doctor assignments for urgent incoming patients based on real-time availability.
- Business Logic: When an emergency request is generated (handleERRequest in the ClinicalViewModel), the system must automatically execute the matchDoctor function. It will query the Staff table to find a Doctor whose specialization perfectly matches the ER request's required specialty.

- Validation: The system must verify that the matched Doctor currently has an active Shift at the ER location, and that their doctorStatus enum is strictly set to AVAILABLE (bypassing any doctors marked as IN_EXAMINATION or OFF_DUTY). The system must also confirm the doctor is within their scheduled start_time and end_time block in the Shifts table.
- System Action: If a valid doctor is found, the system instantly assigns them, changes their doctorStatus to IN_EXAMINATION, and triggers the notifyER method. If no available specialist is found on the current shift roster, the system automatically flags the request in red on the Admin dashboard for immediate manual intervention. In the event of manual intervention, the Administrator can override the assignment to a doctor currently IN_EXAMINATION if their current appointment is nearing its end_time.
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System Trigger	Logic / Filter Applied	Resulting Action
handleERRequest	Doctor.specialization	Specialty Match: Automatically executes matchDoctor to find a specialist matching the ER request.
Availability Filter	DoctorStatus.AVAILABLE	Dispatch Guard: Strictly bypasses doctors currently marked as IN_EXAMINATION or OFF_DUTY.
Successful Match	Doctor.status -> IN_EXAMINATION	Assigns the doctor to the case and triggers the notifyER method.
Match Failure	UI Alert (Red Flag)	Manual Override: Admin may manually assign a doctor who is nearing the end_time of their current examination.

Category	Status Name	Description / Logic Trigger
Staff Availability	is_available (Boolean)	True: Staff is clocked in/on shift. False: Staff is off-duty.
Doctor State	AVAILABLE	Doctor is on shift and ready for an appointment or ER dispatch.
Doctor State	IN_EXAMINATION	Doctor is currently with a patient; bypassed by the "Automated ER Dispatch".
Doctor State	OFF_DUTY	Doctor is not on a scheduled shift; blocked from all assignments.
Shift Lifecycle	SCHEDULED	Initial state when an Admin creates a shift roster.
Shift Lifecycle	ACTIVE	The shift is currently happening; triggers is_available = true for the staff member.
Shift Lifecycle	COMPLETED	The shift has ended; prevents any further modifications or assignments.
Appointment	Scheduled	The appointment is booked and awaiting the start time.
Appointment	Finished	The consultation is over; updates the Appointments table status.

Doctor

1. Process medical evaluations and diagnoses - as a doctor, I need to process incoming symptom data and store the resulting medical diagnosis that I create. I can do this by navigating to the Medical Evaluations window. I need to be able to:
    - Generate a new diagnosis record containing the symptoms identified during the consultation. Additionally, I must include a medications field (meds_list) and doctor's notes (doctor_notes containing dosage and frequency instructions) so that the Prescription module can process them(the Pacients team).
    - Retrieve and display a list of my past medical evaluations and the specific diagnoses I have issued.
    - Edit an existing diagnosis or append supplementary medical notes by selecting a past record and pressing an "Edit Diagnosis" button.
    - Remove an existing diagnostic pressing the “Delete Diagnostic” button.
2. See schedule - as a doctor, I need to see my standard appointments and work-shifts so I know when and where I am working. I can do this by navigating to the Doctor Schedule calendar. I need to be able to:
    - View a calendar component displaying all of my upcoming scheduled medical appointments.
    - Generate and display a daily or weekly shift roster showing when I am on duty.
    - View the specific location for my shift, such as the ER or a specific Clinic Room.
    - See the exact time blocks for my shifts, represented by the Shift Start Time and Shift End Time.
3. Shift Swap -Description: As a Medical Staff member, I need to request a shift swap with a colleague so I can manage my personal schedule.
    - I navigate to the "My Schedule" view.
    - I select a future shift and press the "Request Swap" button.
    - I select a colleague from a list (filtered by the same department/role).
    - The system validates if the colleague is already working during that time.
    - Upon confirmation, a notification is sent to the colleague.

4. Staff hangouts:

    - As a doctor, I can navigate in a “hangouts” tab where I can create a new hangout.
    - A hangout must have a title (5 – 25 characters), a description (max 100 characters), a date (at least 1 week away from the day it is created) and a maximum amount of staff that can join in the hangout (amount selectable in a dropdown table). To create a hangout I need to have no appointments or medical evaluations scheduled for that day. Same goes for joining one.
    - Staff that participate in a hangout will have a small bonus (3-5%) added to their salary to encourage teambuilding.
    - A doctor can join any amount of hangouts he possibly can (given his schedule permits it), but the bonus is limited to one/month.

5. Automated Doctor's Fatigue Monitoring & Clinical Lockout:

- As a doctor, I need the system to monitor my active working hours in real-time to prevent medical errors caused by exhaustion.
- The system continuously calculates the elapsed time since the Shift Start Time of the doctor's currently ACTIVE shift. It also aggregates total hours worked within the current 24-hour window by querying the Shifts table for any COMPLETED shifts.
- Before opening the "Medical Evaluations" window or starting a "Generate Diagnosis" action, the system must verify that the doctor has not exceeded 12 consecutive hours of duty .
- What is must do: If a fatigue threshold is reached, the system instantly disables the "Generate Diagnosis" submission form and displays a high-priority warning: *"Clinical Lockout: Maximum duty hours exceeded. Please hand over active cases."* Simultaneously, the system triggers a Fatigue Intervention Required alert on the Administrator's dashboard, highlighting the doctor's status in red for immediate shift reassignment.

6. Automated Medical History Conflict Check

- As a doctor doing and evaluation , I need the system to automatically check the current diagnosis and medication against the patient's historical record to identify a potential risk.
- When a doctor begin entering a new diagnosis, the system must perform a SQL Select to retrieve the DiagnosisResult and MedList from all previous entries in the Medical_Evaluation table associated with the respective PatientID.
- The system parses the historical data form using specific keyword („Allergy" , „Adverse Reactions") and checks if the current med_lists contains any drug that previously caussed an issue for this pacient.
- If a potential conflict is detected, the system highlights the doctor_notes field in yellow and display a „History Alert" pop-up/banner. The doctor then is required to check a „Assumed Risk " box in the UI before the "Generate Diagnosis " button become active , to ensure the history was checked and reviewed.

7. Automated doctor's salary computation

As a doctor, I can see my salary for the last month, computed automatically by the system, based on the following conditions:

- The standard salary for an hour of work is N Lei, and it is multiplied by the number of hours I had on a shift.
- If the shift was at night, the salary is increased with 20%
- If my shift in weekend, it is again increased (on Saturday with 15%, on Sunday with 25%)

- The system adds these results from each shift from the last month, resulting in the INITIAL SALARY
- AFTER INITIAL SALARY IS COMPUTED, if my specialization is harder (e.g., Surgery doctor works harder than ER doctor), my salary is increased with S% (depending on specialization)
- Depending how many years of experience I have, the salary is again increased for each year with 2%. This is the FINAL SALARY

Pharmacist

1. See schedule - as a pharmacist, I need to see my work-shifts and pharmacy

rotations. I can do this by navigating to the Pharmacy Schedule view. I need to be able to:

- View a daily or weekly shift roster displaying when I am scheduled for work.
- See my specific shift assignments for pharmacy rotations.
- View the exact duration of my shift, clearly displaying the Shift Start Time and Shift End Time.
- See the current status of my scheduled shifts, such as "Scheduled", "Active", or "Completed".

2. Vacation Request:

As a pharmacist, I need to submit time-off or vacation requests so the system prevents me from being scheduled when I am unavailable.

- Logic: I navigate to a "Vacation Window" form and select a Start Date and End Date.'
- System Action: Once approved (or automatically logged), the system updates my profile for that specific date range.
- Administrator Block: If the Administrator attempts to "Create New Shift" for me during those approved dates, the system throws an integrity error and forces them to choose an available pharmacist instead.

3. Automated pharmacyst's salary computation

As a pharmacyst, I can see my salary for the last month, computed automatically by the system, based on the following conditions:

- The standard salary for an hour of work is N Lei, and it is multiplied by the number of hours I had on a shift.
- If the shift was at night, the salary is increased with 20%
- If my shift in weekend, it is again increased (on Saturday with 15%, on Sunday with 25%)
- The system adds these results from each shift from the last month, resulting in the INITIAL SALARY

- AFTER INITIAL SALARY IS COMPUTED, then it is increased with M%, depending on how many medicines I sold last month
- Depending how many years of experience I have, the salary is again increased for each year with 2%. This is the FINAL SALARY